A REMARK ON OPENAI FAMILY 108: THE CUBIC CONSTANT FOR $\overline{\mathrm{dc}}(\mathrm{per}_m)$

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Abstract. OpenAI's manuscript *A cubic lower bound for border determinantal complexity of the permanent* (family 108 of `github.com/openai/math`, commit `adc7f124`) proves $\overline{\mathrm{dc}}(\mathrm{per}_m) \geq m^3/(5\,529\,600\,e)$ for $m \geq 1408$. We record three facts. (1) A short lemma ("box lemma") improves the manuscript's length bound length $Z \geq \binom{q+2}{2}$ to length $Z \geq (q+1)^2$ and removes the hypothesis $q \leq r-1$. Together with re-chosen parameters this gives $\overline{\mathrm{dc}}(\mathrm{per}_m) \geq m^3/(125\,500\,e)$ for all $m \geq 1408$, and $m^3/((118\,800+o(1))e)$ asymptotically: 44× resp. 46.5× better. Parameter choice alone accounts for about 21×/23×; the box lemma for the remaining factor 2.1×/2.0×. (2) The gradient locus contains explicit block-triangular families, valid in every characteristic, whose codimension is asymptotically $(\frac{1}{2}-\beta^2)k^2$ with $\beta = r/k$. This is smaller than $(2q+2)^2$, the codimension of the rank family. So the latter is not the true codimension, and two natural strengthening lemmas are false. (3) Consequently no proof inside the manuscript's pipeline (selector, smooth linear section, smooth-hypersurface theorem) can give a uniform constant better than $1/((33\,065+o(1))e)$, i.e. more than 167× the stated one. The construction and all main arguments are OpenAI's.

1. Setting

We use the notation of the manuscript. K is algebraically closed, $\theta_1, \dots, \theta_k \in K$ are distinct, $D = \mathrm{diag}(\theta_i)$, $0 \leq q \leq (k-2)/2$, $r = k - 2q \geq 2$, $c = r - 1$, and

$$G_D(X) = [s^q t^q] \det(X + sI + tD), \qquad \Gamma_D = \{X : \nabla G_D(X) = 0\} \subset \mathrm{Mat}_k(K).$$

For fixed X let $M = sI + tD + uX$, $F_X = \det M$, $C_X = V(F_X) \subset \mathbb{P}^2$, and let $Z = Z_X \subset C_X$ be the finite scheme cut out by the entries of $\mathrm{adj}\, M$ (`adjugate.tex` L123–137). Let $I_Z \subset S = K[s,t,u]$ be its saturated homogeneous ideal.

The manuscript's chain is: $\nabla G_D(X) = 0 \iff [s^q t^q u^c]\, \mathrm{adj}\, M = 0$ (`adjugate.tex` L261–270, no hypothesis on q); length $Z \geq T := \binom{q+2}{2}$ (Prop. `adjugate-length`, L250–309, which uses $q \leq c$ at L291); length $Z \leq \delta(C_X)$ (Prop. `conductor-length`); hence $\dim \Gamma_D \leq k^2 + k - T$ (`dimension.tex` L10–27, L353–366). With $\Delta = T - k$, characteristic transfer, a smooth linear section of dimension d with $2 \leq d < k^2$, $d - 1 < \Delta$, and the selector $m \geq 11k - 2q$ (`transfer.tex` L218–309, `selector.tex` L55–58), Theorem `smooth-border` gives

$$\overline{\mathrm{dc}}(\mathrm{per}_m) \;\geq\; \frac{(r-1)(d-1)}{4e}. \tag{1}$$

The manuscript takes $q = \lfloor (k-1)/3 \rfloor$, $d = \lfloor k^2/100 \rfloor$, $k = \lfloor m/11 \rfloor$ (`consequences.tex` L8–38).

Related work. Two follow-ups bound other quantities and leave the cubic constant 5 529 600 unchanged. Guo [2] (v2, October 8, 2026) moves the smooth slice of family 108 to circuits with division and obtains $L_{\mathrm{div}}(\mathrm{per}_n) \geq (n^2/86400)\log_2(n/48)$ ($n \geq 1408$), assuming the manuscript's Prop. 7.3; from the Lean declarations alone he obtains $(n^2/119790)\log_2(n/44)$ ($n \geq 1936$). Xie [3] (October 2, 2026) proves $\overline{\mathrm{dc}}(\mathrm{per}_n) = \Omega(n^2 \log n)$, explicitly $\geq n^2(\log_2 n - 5)/(64e)$ for $n \geq 64$.

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Paper versus Lean. Guo also points out that the Lean slice size $D = \lfloor \binom{q+2}{2}/18 \rfloor$ differs from Prop. 7.3. We confirm this. The Lean development (`Main.lean`, `critical_dimension_effective`) proves only $\dim \Gamma_D \le k^2 - \lfloor T/18 \rfloor$, where the paper's Prop. 6.1 has $k^2 + k - T$. It reaches the same constant $5\,529\,600$ through tighter auxiliary estimates ($r - 1 \ge 0.33k$, $D - 1 \ge k^2/330$, $m \le 11.1k$). Our baseline and all our constants follow the paper: Prop. 6.1, $d \le \Delta$, and $5\,529\,600 = 12^3 \cdot 3200$ from `consequences.tex`. The number is the same in both sources, so the ratios reported below are unaffected. However, our constants rest on the paper's Prop. 6.1 and are not Lean-certified. If one uses only the Lean dimension bound, re-chosen parameters give $c^{-1} \approx 4.2 \cdot 10^6 e$ (1.3×).

2. The box lemma

Lemma 1 (Box lemma)**.** *Let $0 \le q \le (k-1)/2$ and $X \in \Gamma_D$. Then* $\operatorname{length} Z_X \ge (q+1)^2$.

Proof. Put $\Lambda = [s^q t^q u^c]$, a linear functional on S_{k-1}.

Step 1. By Lemma `adjugate-generation` (`adjugate.tex` L139–248, no hypothesis on q), every form of degree $k-1$ in I_Z is a constant linear combination of entries of $\operatorname{adj} M$. These entries are killed by Λ because $X \in \Gamma_D$. Let $P \in (I_Z)_d$ with $d \le k-1$. For every monomial Q of degree $k-1-d$ we have $PQ \in (I_Z)_{k-1}$, so $\Lambda(PQ) = 0$. In other words, no monomial dividing $s^q t^q u^c$ occurs in P. Thus

$$(I_Z)_d \subset (s^{q+1}, t^{q+1}, u^{c+1})_d \qquad (d \le k-1).$$

This is coefficient extraction, not differentiation, so it holds in every characteristic, including 2. It is the computation of `adjugate.tex` L283–297, applied to all degrees and not only to degree q.

Step 2. Let $B = (s^{q+1}, t^{q+1}, u)$. For $d \le 2q \le k-1$, Step 1 gives $(I_Z + (u))_d \subset B_d$. For $d \ge 2q+1$, every monomial of degree d in s, t is divisible by s^{q+1} or t^{q+1}, so $B_d = S_d$. Hence $I_Z + (u) \subset B$ in every degree.

Step 3. Z does not meet the line $u = 0$. The points of C_X on it are $P_i = [-\theta_i : 1 : 0]$, and there the diagonal adjugate entry $\prod_{j \ne i}(\theta_j - \theta_i)$ is nonzero (`adjugate.tex` L81–87). So u lies in no associated prime of S/I_Z and is a nonzerodivisor. Then $H_{S/(I_Z+u)}(d) = H_{S/I_Z}(d) - H_{S/I_Z}(d-1)$, and summing gives

$$\operatorname{length} Z = \dim_K S/(I_Z + (u)) \;\ge\; \dim_K S/B = (q+1)^2. \qquad \square$$

Remark 2. The manuscript's argument extracts only the injectivity of $S_q \to H^0(\mathcal{O}_Z(q))$, which gives $\dim S_q = \binom{q+2}{2}$. Step 3 turns the whole Hilbert function of the Artinian reduction into length. The lemma is short; we regard it as a small improvement, not as new technique.

Replacing T by $(q+1)^2$ in `dimension.tex` gives $\dim \Gamma_D \le k^2 + k - (q+1)^2$ for every $0 \le q \le (k-2)/2$. The Severi bound, the conductor bound, and the fibre bound do not involve q.

3. Constants

With $\Delta = (q+1)^2 - k$ we take $d = \Delta$ (so $d - 1 < \Delta$), which needs $2 \le \Delta < k^2$, $r \ge 2$, and $11k - 2q \le m$. We maximise $(r-1)(d-1)$ in (1). Write $c^{-1} = 4m^3/((r-1)(d-1))$ in units of e.

Theorem 3. *For every integer $m \ge 1408$, $\overline{\mathrm{dc}}(\mathrm{per}_m) \ge \dfrac{m^3}{125\,500\, e}$. Moreover $\overline{\mathrm{dc}}(\mathrm{per}_m) \ge \dfrac{m^3}{(118\,800 + o(1))\, e}$ as $m \to \infty$. The same bounds hold for* $\mathrm{dc}(\mathrm{per}_m)$.

Proof sketch. Asymptotics. Put $q = \gamma k$. Then $m \approx (11-2\gamma)k$ and $(r-1)(d-1) \approx (1-2\gamma)\gamma^2 k^3$. The function $(1-2\gamma)\gamma^2/(11-2\gamma)^3$ is maximised at $\gamma = 11/32$, with value $1/29\,700$. So $c^{-1} \to 4 \cdot 29\,700 = 118\,800$.

Tail. Take $k = \lfloor 32(m-2)/330 \rfloor$ and $q = \lfloor 11k/32 \rfloor$. Then $11k - 2q \le m$, $r - 1 \ge 10k/32 - 1$, and $d - 1 \ge (11k/32)^2 - k - 1$. Substituting $k \ge 32(m-2)/330 - 1$ gives an explicit rational function $g(m) \ge c^{-1}(m)$. The numerator of g' has no real root above 65 (exact computation), so g decreases. Also $g(200\,000) = 118\,893$.

Range $1408 \le m \le 200\,000$. For each m we search (k, q) near the optimum. The largest value of $c^{-1}(m)$ found is 125 422.2, at $m = 1419$. See `scripts/independent/uniform_indep.py`. □

	uniform, $m \ge 1408$	$m = 5000$	$m \to \infty$
stated (`introduction.tex` L48)	5 529 600	5 529 600	5 529 600
manuscript's lemma, re-chosen parameters	267 955	245 793	238 328
box lemma, re-chosen parameters	125 422	120 334	118 800
gain from parameters only	20.6×	22.5×	23.2×
gain from box lemma	2.14×	2.04×	2.006×
total	44.1×	46.0×	46.5×

TABLE 1. Values of c^{-1}/e. "Re-chosen parameters" means $q \le (k-1)/3$, $d = \Delta$, and k maximal subject to $11k - 2q \le m$. Of the box-lemma factor 2.006, the change $\binom{q+2}{2} \to (q+1)^2$ contributes 2.000, and dropping $q \le c$ contributes 1.003.

Most of the loss in the stated constant comes from the manuscript's deliberately crude estimates $d \approx k^2/100$ against $\Delta \approx k^2/18$, $k \ge m/12$, and $r - 1 \ge k/4$ against $k/3$. These alone are worth $\approx 23\times$.

4. The true codimension and the limit of the method

Proposition 4 (Block families)**.** *The partials of G_D are K-linear combinations of the* almost-principal $(r-1)$*-minors of X, that is, minors with row set $U \cup \{a\}$ and column set $U \cup \{b\}$. Let X be block upper triangular for some ordered partition of the coordinates into m' blocks, and let the diagonal blocks have ranks ρ_i with $\sum \rho_i + m' \le r - 1$. Then every such minor vanishes, so $X \in \Gamma_D$. This holds for every field, every characteristic, and every θ.*

Proof. $\partial G_D/\partial X_{ij} = [s^q t^q] \operatorname{adj}(X + sI + tD)_{ji}$. Expanding along the diagonal $(s + t\theta_l)$ entries gives a sum over $2q$-sets S of a polynomial in θ times the minor of X with rows $[k] \setminus (S \cup \{i\})$ and columns $[k] \setminus (S \cup \{j\})$. For the vanishing, apply the generalized Laplace expansion along the row blocks: $\det X[I, J] = \pm \sum \prod_p \det X[I_p, C_p]$, where C_p consists of columns in blocks $\ge p$. A term is nonzero only if C_p contains at most ρ_p columns of block p. Process the blocks in order. Since $|I_p|$ and $|J_p|$ differ in at most two blocks and by at most one, at most $m' - 1$ columns are assigned off the diagonal. So $\sum_p \rho_p \ge (r-1) - (m'-1)$, which contradicts the hypothesis. □

Take one block of size $n_1 \approx 2\beta k$ and rank $r - 1 - m'$, and let the remaining $m' - 1 \to \infty$ blocks have rank 0 and equal sizes. The family has codimension $(\frac{1}{2} - \beta^2 + o(1))k^2$, where $\beta = r/k$. The case $m' = 1$ is the rank-$\le r - 2$ family, of codimension $(2q+2)^2 \approx (1-\beta)^2 k^2$. Since $(1-\beta)^2 - (\frac{1}{2} - \beta^2) = 2(\beta - \frac{1}{2})^2$, the block families are strictly larger for $\beta \ne \frac{1}{2}$. Two explicit instances, each verified by exact computation over $\mathbb{Q}$ and $\mathbb{F}_{2^{20}}$:

- $k = 9$, $q = 3$: $X = \left(\begin{smallmatrix} 0 & Y \\ 0 & 0 \end{smallmatrix}\right)$ with $4 + 5$ blocks has codimension $61 < 64 = (2q+2)^2$.
- $k = 22$, $q = 7$: blocks $(16, 6)$, ranks $(5, 0)$, codimension $253 < 256$.

Hence a lemma of the form $\operatorname{codim}\Gamma_D \geq (2q+2)^2 - O(1)$ is false. The natural intermediate lemma $\dim\{X : \text{length}\, Z_X \geq \ell\} \leq k^2 + k - 2\ell$ is also false. For $k = 16$ and $X = \left(\begin{smallmatrix} 0 & Y \\ 0 & 0 \end{smallmatrix}\right)$ with 8×8 blocks, every adjugate entry has order $\geq k-2$ at $[0:0:1]$. So length $Z_X \geq \binom{15}{2} = 105$ for every Y, and the local length is exactly 105. This family has dimension $64 > 62$.

Proposition 5 (Ceiling)**.** *Consider any proof that applies Theorem* `smooth-border` *to a d-dimensional linear section of G_D, for some diagonal D, and embeds it in per_m by the selector ($m \geq 11k - 2q$). Then $d \leq \operatorname{codim}\Gamma_D$, by the projective dimension theorem and Euler's identity over $\mathbb{C}$. Proposition 4 therefore gives*

$$\frac{(r-1)(d-1)}{4e} \;\leq\; \frac{(1+o(1))\,m^3}{C_{\mathrm{ceil}}\,e}, \qquad C_{\mathrm{ceil}} = \min_{0<\beta<1} \frac{4(10+\beta)^3}{\beta \min\{(1-\beta)^2, \frac{1}{2} - \beta^2\}} = 33\,065.4\ldots,$$

attained at $\beta = (\sqrt{601} - 1)/60 \approx 0.392$. So no uniform constant from this pipeline beats $1/((33\,065 + o(1))e)$, a factor $167\times$ over the stated constant. The box lemma is within $3.59\times$ of this ceiling.

At finite m the per-m ceiling is slightly weaker: $30\,872e$ at $m = 1408$ ($179\times$) and $31\,858e$ at $m = 5000$ ($174\times$). The ceiling depends on the premises stated. It does not cover initial forms at non-zero points, other coefficients $[s^a t^b]$, or other selectors.

The true codimension is therefore in $\big[(q+1)^2 - k,\ \min\{(2q+2)^2,\ \text{block-family codim}\}\big]$, asymptotically $[\frac{1}{4}(1-\beta)^2,\ \frac{1}{2} - \beta^2]\,k^2$.

Conjecture 6 (A*, open)**.** *For $\beta = r/k$ fixed in $(\frac{1}{4}, \frac{1}{2})$, $\operatorname{codim}\Gamma_D \geq (\frac{1}{2} - \beta^2 - o(1))k^2$ for generic θ; that is, the block families of Proposition 4 are asymptotically extremal. If true, the pipeline would reach $c^{-1} \to 33\,065e$.*

We have not checked this conjecture beyond small cases, and we do not know whether families with several ranked blocks do better.

5. Computational certificates

All scripts are in `scripts/`, with outputs in `scripts/expected_output/`. `box_indep.sage` checks steps 1–3 of Lemma 1 for $k \leq 9$ over $\mathbb{Q}$, $\mathbb{F}_{1000003}$ and $\mathbb{F}_{2^{10}}$, including cases with $q > c$. `grad_check.sage` checks the counterexamples. It uses the characteristic-free identity $\operatorname{adj}(sI - B) = \sum_j s^{k-1-j} \sum_{i \leq j} c_i B^{j-i}$, which it cross-checks against brute-force differentiation. `minor_span.sage` checks the almost-principal-minor span for $k \leq 7$. `lemmaB_check.sage` computes the local length 105 exactly over $\mathbb{Q}$.

References

[1] OpenAI, *A cubic lower bound for border determinantal complexity of the permanent*, September 24, 2026, `github.com/openai/math`, `preprints/A-cubic-lower-bound-for-border-determinantal-complexity-of-the-permanent-September-24-2026`, commit `adc7f1241b42`.

[2] J. Guo, *An $n^2 \log\log n$ lower bound for permanent circuits with valid division*, arXiv:2609.29568, v2, October 8, 2026.

[3] Y. Xie, *Smooth slices and superquadratic border determinantal complexity of the permanent*, Zenodo, record 23108131, October 2, 2026.

Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.